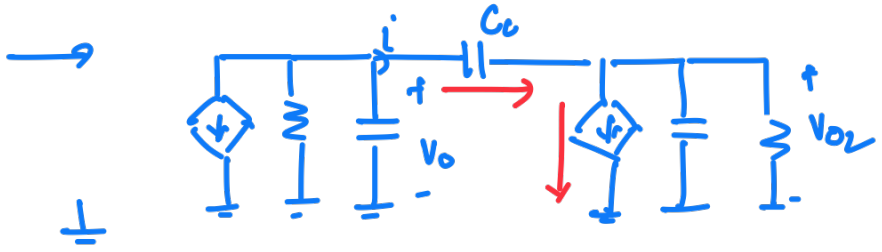
Lecture #19

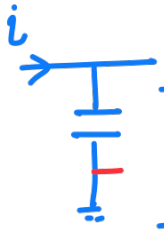
$\frac{N(s)}{D(s)} \rightarrow \frac{(s-\omega_z)}{(s+\omega_p)}$



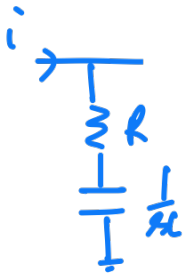
$s+a+b$
>0 <0

$$N(s) = \left(1 - \frac{sC_c}{g_m}\right) = \frac{1}{g_m}\left(g_m - sC_c\right)$$

$$\omega_z = \frac{g_m}{C_c}$$

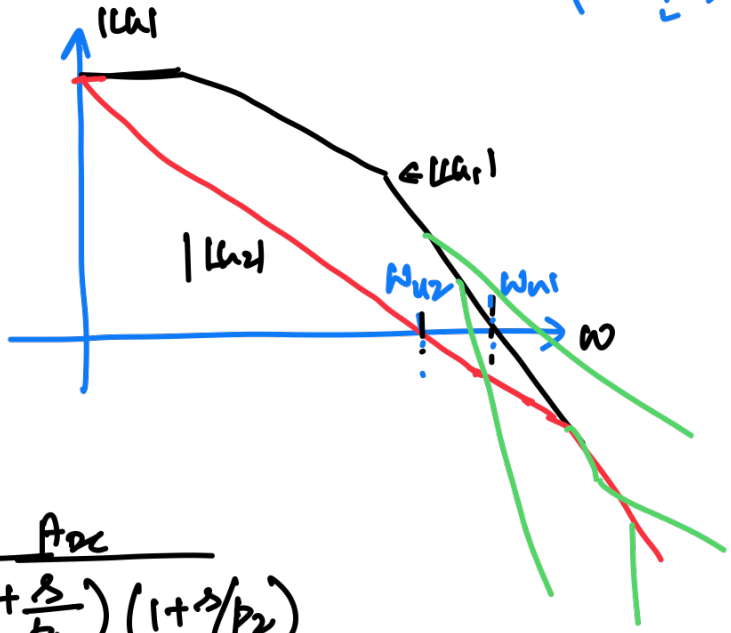


$$V_c = i \times \frac{1}{sC_c}$$



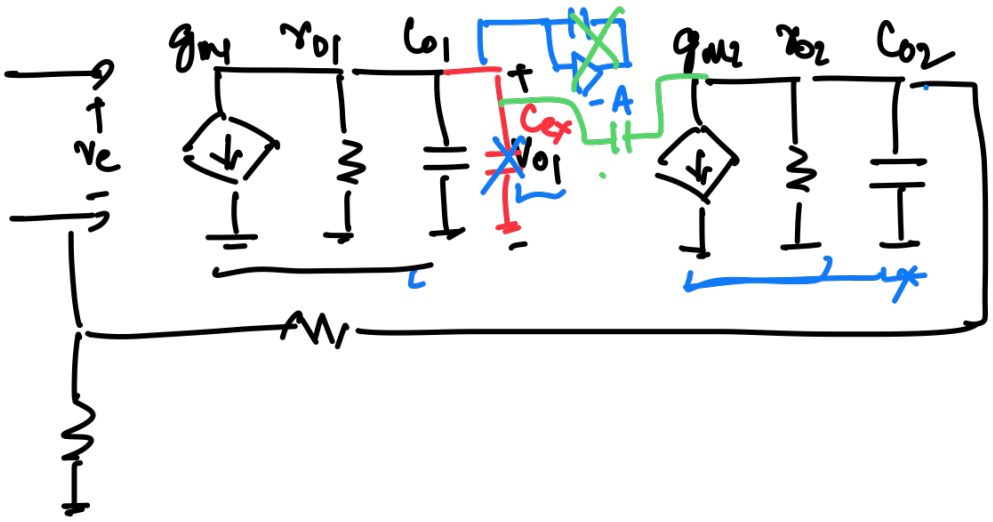
$$V_c = i\left(R + \frac{1}{sC_c}\right) = \frac{i}{sC_c}\left(1 + sRC_c\right)$$

$$H(s) = \frac{A}{as^2 + bs + c} \approx \frac{A}{(s+p_1)(s+p_2)}$$

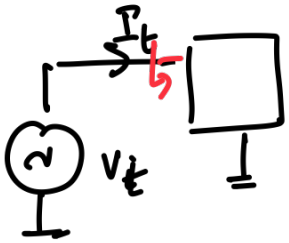


$$L_1 = \frac{A_{DC}}{\left(1 + \frac{s}{p_1}\right)\left(1 + s/p_2\right)}$$

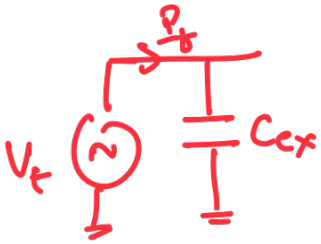
$$|L_1(\omega_u)| \approx \frac{A_{DC}}{\left(\frac{\omega_u}{p_1}\right)\left(\frac{\omega_u}{p_2}\right)} \quad ; \quad \omega_u \gg p_1, \ \omega_u \gg p_2$$



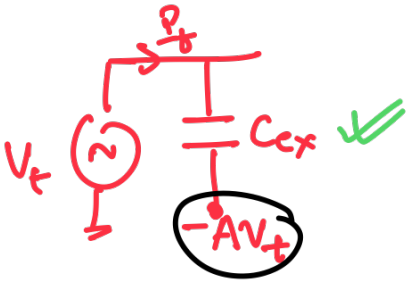
"Dominant pole Comp.'r"



$$Z = \frac{V_t}{I_t}$$



$$Z = \frac{V_t}{I_t} = \frac{1}{sC_{ex}}$$

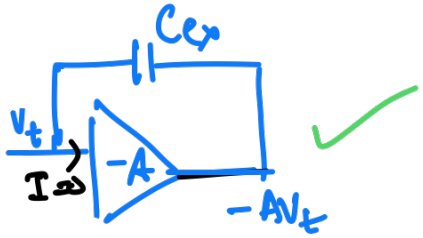


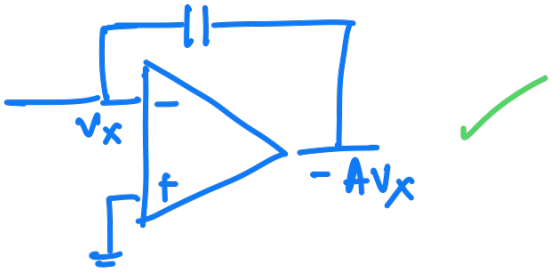
$$I_t = \frac{V_t - (-AV_t)}{1/sC_{ex}}$$

$$= V_t \, sC_{ex}(1+A)$$

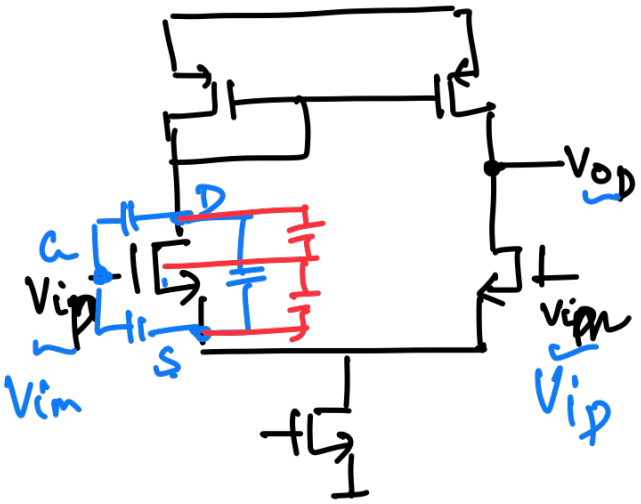
$$= V_t \, sC_y$$

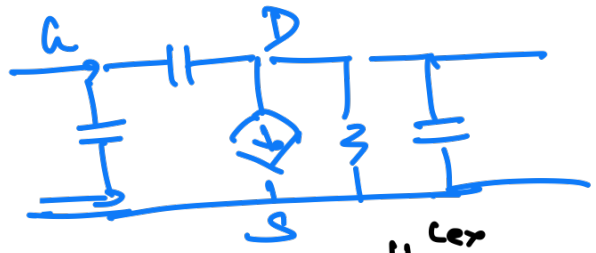
$$C_y = C_{ex}(1+A)$$

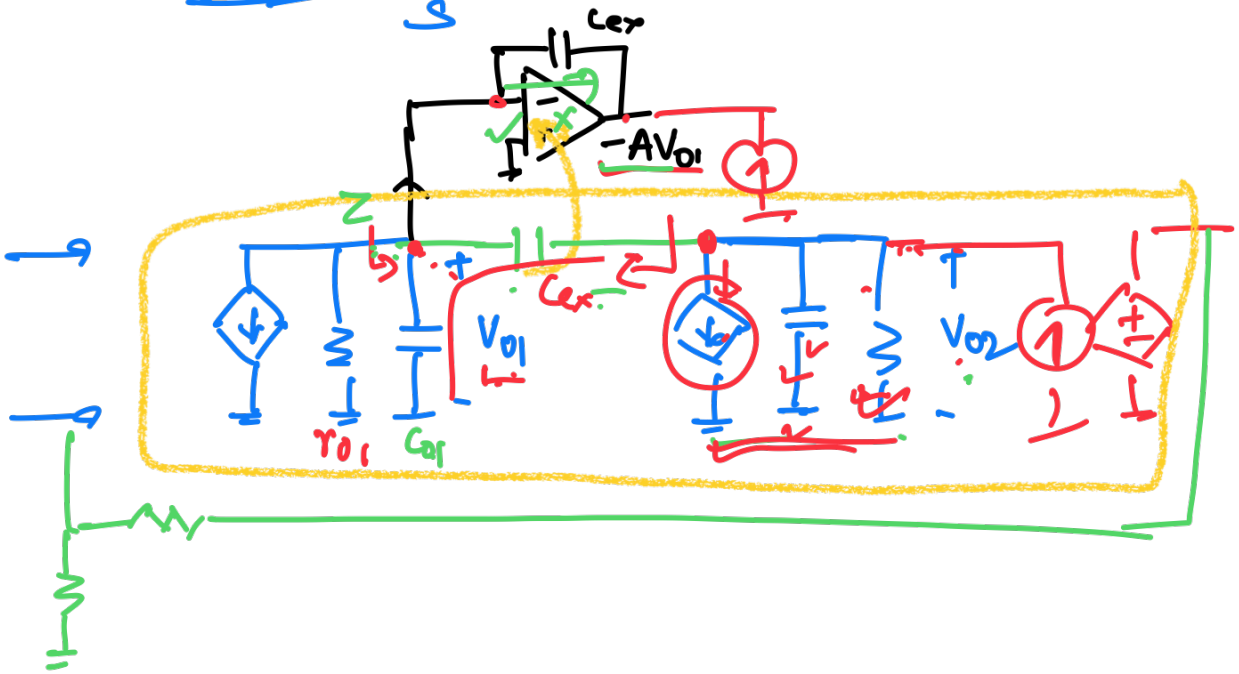




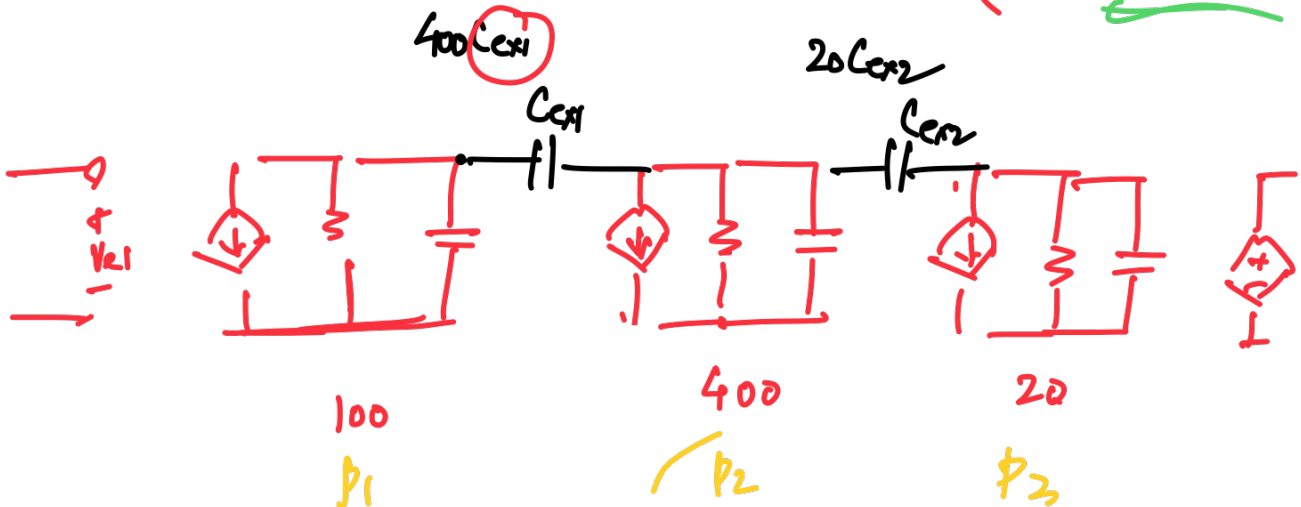
LH = _____







$$V_{o2} = \frac{-g_{m2} r_{o2}}{(1+sC_{o2}r_{o2})} \qquad V_{o1} = \frac{-AV_{o1}}{(1+sC_{o2}r_{o2})}$$



$$\omega_n = \sqrt{A_{DC}\, p_1 p_2}$$

$$\omega_n = 1.1 p_2$$

$$= 1.0\ldots01 f_c$$

$$= 0.999\ldots8p_1 = 1$$

$$|L_{u2}(j\omega_{n2})| \cong \frac{A_{DC}}{\frac{\omega_{n2}}{p_1}}$$

$$L_u = \frac{A_{DC}\ (1 - s/\omega_z)}{(1+s/\omega_{p1})\ (1+s/\omega_{p2})}$$

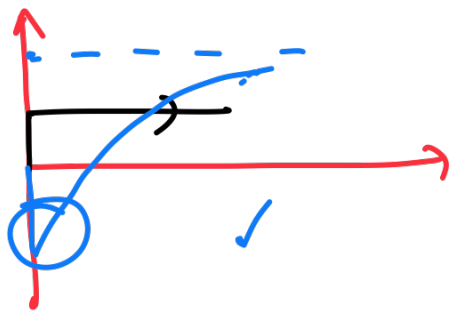
$$\angle L_u = -\tan^{-1}\left(\frac{\omega_n}{\omega_{p1}}\right) - \tan^{-1}\left(\frac{\omega_n}{\omega_{p2}}\right) - \tan^{-1}\left(\frac{\omega_n}{\omega_z}\right)$$

$$\phi_m = \angle L_u(\omega_n) - (-180^\circ)$$

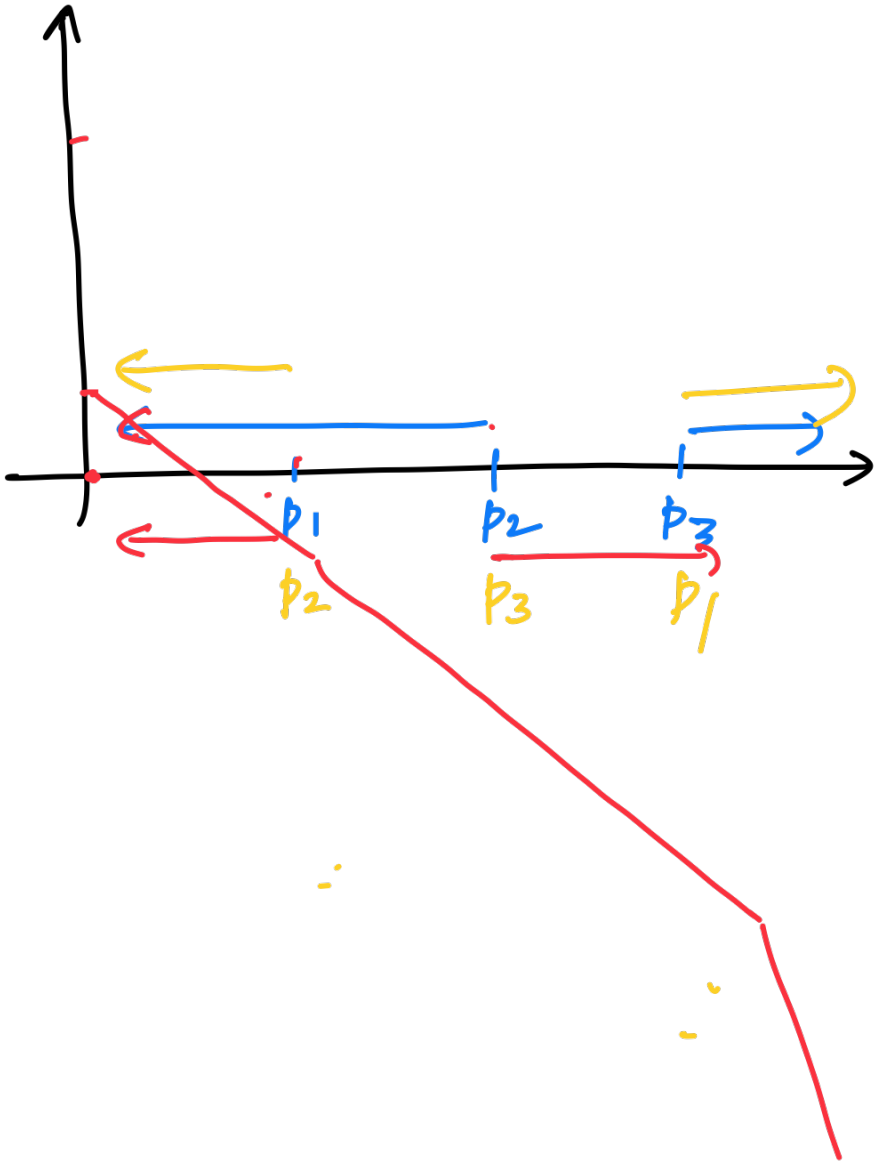
$$= 180^\circ - \tan^{-1}\left(\frac{\omega_n}{\omega_{p1}}\right) - \tan^{-1}\left(\frac{\omega_n}{\omega_{p2}}\right) - \tan^{-1}\left(\frac{\omega_n}{\omega_z}\right) - \tan^{-1}\left(\frac{\omega_n}{\omega_{p3}}\right)$$

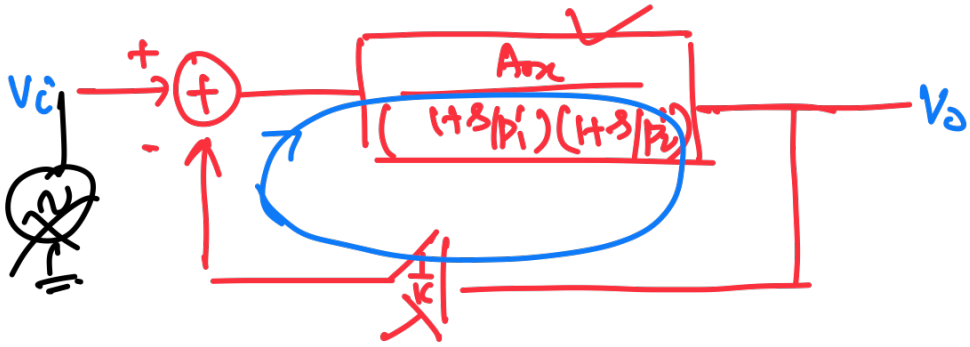
10°

$\phi_m = 45^\circ$



- Minimum physical capacitance. $ Vs area.
- Maximum unity gain bw. 100 MHz.
- Desired phase margin. 70°
- Minimum power budget.
- Minimum noise. 1μV rms noise.

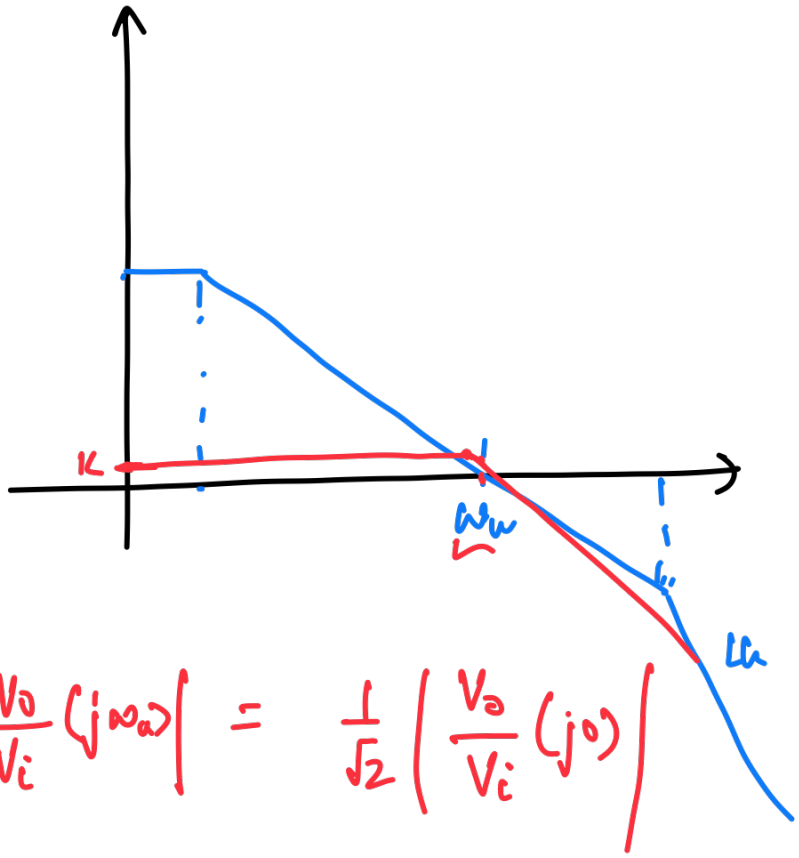




Loop Gain, Phase margin ana. is independent of input.

$$\frac{V_o}{V_i} = \frac{A(s)}{1 + A(s)H(s)} = \frac{A(s)}{1 + LA} = K\,\frac{LA}{1 + LA}$$

$$= \frac{K}{1 + \frac{1}{LA}}$$



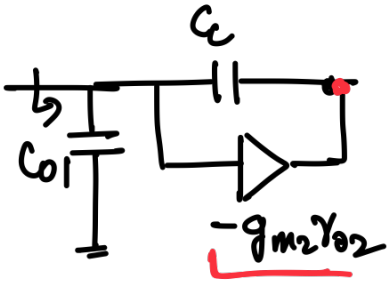
$LA(j\omega_u) = 1 \angle 90^\circ$

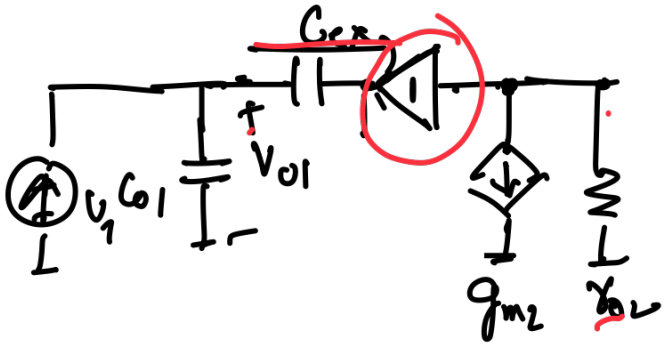
$\left|\frac{K}{1 + \frac{1}{LA}}\right| = \frac{K}{\sqrt{2}}$

$20\log\left(\frac{1}{\sqrt{2}}\right) = 3$

$$\left|\frac{V_o}{V_i}(j\omega_u)\right| = \frac{1}{\sqrt{2}}\left|\frac{V_o}{V_i}(j0)\right|$$

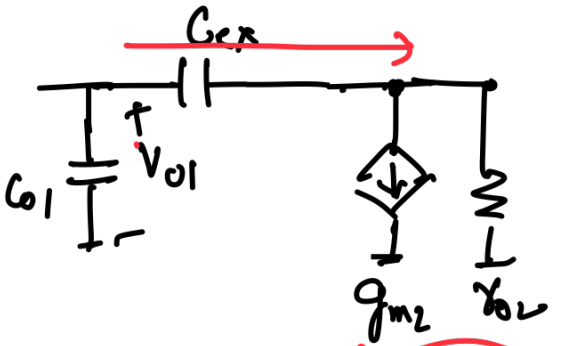
$$p_1' = \frac{}{C_{01} + C_c(1 + g_{m2}r_{o2}) + \cdots}$$



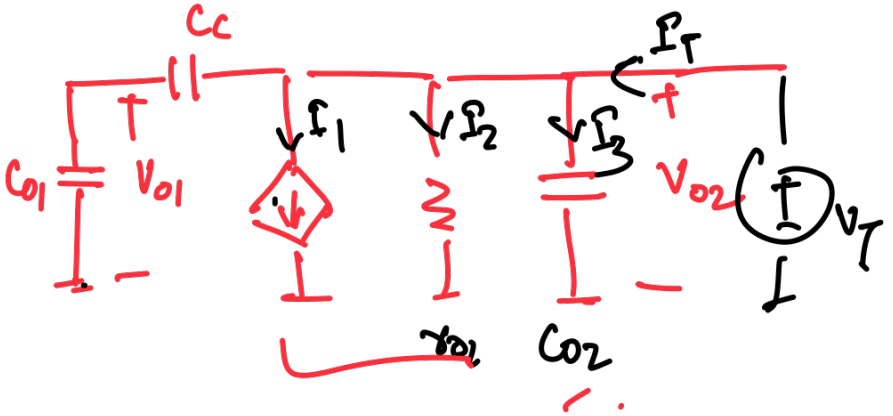


$C_{01} + C_c(1+g_{m2}r_{o2})$

$g_{m2}r_{o2}V_{01}$



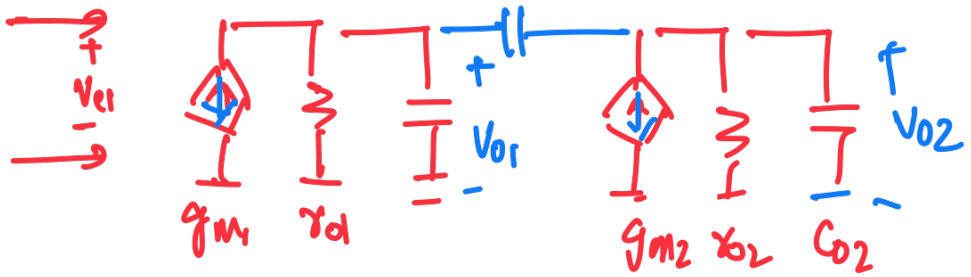
$g_{m2}r_{o2}V_{01}$



$$V_{01} = \frac{C_c}{C_c + C_{01}}$$

$$I_1 = V_T \, g_{m2} \frac{C_c}{C_c + C_o}$$

$$I_2 = \frac{V_T}{r_{o2}}$$



$$\frac{V_{o2}}{V_{o1}} = \frac{g_{m2} r_{o2}}{1 + s C_{o2} r_{o2}}$$